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Response to CMS RFI: Modernizing the Health Technology Ecosystem

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Introduction

Confluent Federal is pleased to submit this response to the Centers for Medicare & Medicaid Services (CMS) Request for Information (RFI) on building a modern, interoperable health technology ecosystem. Confluent delivers the leading platform for data in motion powered by Apache Kafka, enabling secure, real-time data streaming across complex and distributed environments. Our technology is widely adopted across federal agencies and commercial healthcare organizations to modernize legacy systems, reduce latency in care delivery, and empower decision-making with live data.

Within the Centers for Medicare & Medicaid Services' push to modernize and consolidate their vast health technology ecosystem, there lies an opportunity to rethink how data is moved, shared, and acted upon. Many of the frustrations and inefficiencies that currently slow healthcare progress stem from entrenched silos and the disjointed, batch-oriented systems of yesterday. A modern federal health enterprise demands agility, security, and the power to derive actionable insights from fresh data, wherever and whenever it is needed. Confluent's data streaming platform serves as a proven foundation for achieving these goals, weaving together the very capabilities that CMS must consider to make its modernization journey successful.

CMS is tasked not only with overseeing the nation's largest payer system, but with setting a standard for interoperability, privacy, equity, and efficiency across the healthcare continuum. The current landscape is fragmented, with essential insights locked in legacy databases or distributed across multiple clouds, on-premises data centers, and siloed applications. As CMS embarks on the journey to consolidate and modernize, a real-time, event-driven architecture offers a pathway to eliminate data silos and streamline data sharing across domains, while supporting the flexibility demanded by both enterprise-scale operations and specialized programs. A data streaming platform approach, like the one offered by Confluent, enables data to be shared securely and reliably on-prem and in the cloud, under strict schema governance and with shift-left strategies that enrich data and resolve errors early, rather than downstream.

Developers, Data Providers, and Technology Vendors

For those building, integrating, and maintaining the new digital backbone at CMS, flexibility and interoperability are imperative. Event streaming with Confluent allows developers, health IT vendors, and data providers to decouple systems and adopt integration patterns that are modular and robust. Open APIs, FHIR, and HL7 standards

allow development teams to establish seamless pipelines that move data across on-prem, multi-cloud, and legacy environments, all while ensuring mainframe offloading or cloud migration can happen incrementally, without risk to mission continuity.

Using decentralized data streaming and data mesh patterns, teams can bridge both domain-specific and enterprise-wide boundaries, making sure that systems communicate regardless of their original architecture. Confluent's cluster linking feature mirrors events and data across environments, enabling efficient data sharing and modernization with zero downtime. Schema registry, schema validation, data contracts, and metadata tagging offer a new level of transparency and trust in integrations. By enforcing strong data governance at the forefront, teams can discover, share, and quickly onboard reusable data products, while supporting 1-to-many distribution that reduces duplicative pipelines and technical debt. This shift ensures CMS won't be locked into one set of tools or vendors, but will have an extensible, source-of-truth architecture that saves money and accelerates innovation over the long run.

Patients and Caregivers

For the beneficiaries of CMS programs and their caregivers, modernization isn't just a technical exercise—it transforms the patient experience. Imagine a system where the Single Patient View is a reality: as a patient transitions between providers, hospitals, or even at-home care, their records, lab results, device readings, and orders move with them, up to date and denormalized for immediate use. Confluent enables longitudinal patient records and near-real-time aggregation from disparate systems, supporting continuous monitoring—whether via EHRs, IoT devices, or telemedicine platforms.

Telemedicine applications built atop real-time streaming architectures ensure clinicians have immediate access to the most current patient data, while caregivers can rely on event-driven alerts for everything from medication refills to eligibility determinations. This architecture enables actionable triggers for notifications and reminders, improving outcomes by keeping contextual data perpetually current and relevant. AI-powered clinical training environments can be constructed using de-identified, richly annotated streams, giving tomorrow's workforce realistic, compliant datasets to hone their skills.

Providers

Providers—hospitals, clinicians, and coordinating agencies—face mounting documentation burdens, disconnected care workflows, and administrative slowdowns. Event-driven health IT powered by Confluent addresses these challenges by integrating clinical and operational systems in real time. Patient admission, discharge, and transfer

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(ADT) workflows, which historically rely on error-prone batch processing, transform into fluid exchanges where updates ripple across EHR, lab, billing, and pharmacy systems as they happen.

This same architecture supports real-time claims management, dramatically abbreviating adjudication cycles and reducing the effort required to keep provider portals and patient-facing applications updated. Schema validation and strict governance throughout the streaming pipelines enforce data quality, while shift-left stream processing strategies catch errors and standardization issues early. The result is a reduction in duplicative effort, higher trust in the timeliness and accuracy of source-of-truth data, and more time for clinicians to focus on patient care. Advanced use cases, such as real-time sepsis monitoring and AI-driven early interventions in cancer care, become possible as continuous, high-quality streams feed analytics and ML models.

Payers

CMS and payer partners—insurers, accountable care organizations, and program administrators—need new tools to thrive in an era of value-based care. With fully integrated, event-based data pipelines, claims, utilization, and clinical trends are no longer locked in departmental silos. Streaming architectures based on Confluent allow for rapid, responsive calculations of eligibility, risk, and payment. By harnessing change data capture integrations and robust streaming patterns, payers can comply with interoperability mandates, such as CMS interoperability rules, with agility.

Fraud detection too becomes a real-time activity, benefiting from event triggers that highlight abnormal usage patterns or claims, while machine learning integrations can surface emerging threats before they escalate. The ability to combine a Customer 360 approach—bringing together longitudinal and transactional data across a member's journey—enables more accurate risk assessments, proactive outreach, and rapid response to health events, ultimately leading to better health outcomes and cost efficiencies.

Security and Trust

Security and privacy are pillars of any CMS modernization effort, and Confluent's platform's enterprise-grade security is architected for compliance from the ground up. Robust data encryption both in transit and at rest, mutual authentication, role-based access controls, and advanced audit trails all ensure sensitive health information is always protected. Data stream governance provides continuous visibility and enforce compliance with HIPAA, NIST, Zero Trust, PHI/PII, and federal regulatory frameworks. Multi-tenancy support allows CMS to share infrastructure securely across programs and partners while isolating sensitive datasets as needed, reducing duplication and protecting privacy without locking data away in isolated silos.

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Real-time event triggers further enhance security posture by alerting operations teams to anomalies as they occur, supporting zero-trust architectures and integrations with SIEM tools for ongoing monitoring and incident response. By combining shift-left methodologies, enterprise security, and schema enforcement, CMS can drive data integrity and guard against downstream data breaches or errors before they propagate.

Outcomes and Equity

The end goal of modernization, beyond mere efficiency, is to improve healthcare outcomes and advance equity. Confluent empowers CMS to create a decentralized, governed ecosystem of real-time data products, where accurate denormalized datasets can be composed for individuals, cohorts, and populations. A data mesh approach unlocks cross-domain collaboration and analytics, so population health managers can spot disparities, react to events as they occur, aggregate results in real time, integrate AI/ML/GenAI workflows, or rapidly deploy new services in response to emerging health threats.

The ability to provide contextual, actionable data to everyone—from clinicians to community health workers—ensures interventions are timely and relevant, closing gaps in care. Cost savings accrue as technical debt is systematically reduced, duplicative ETL and integration work is eliminated, and durable, extensible architectures allow federal programs to reinvest in care improvement, research, innovation, and core patient and member services.

Potential Pilot Ideas

To jumpstart this vision, CMS could consider piloting an Event Streaming Reference Architecture that demonstrates longitudinal patient records, real-time ADT synchronization, and secure streaming from wearables and home devices, all unified under standardized event-driven APIs. Further pilots might focus on feeding AI models to identify high-risk patients in real time, implementing fraud detection through event-based anomaly monitoring, or building data marketplaces where securely managed data products support public-private innovation.

By starting with such pilots, CMS can validate technical approaches, quantify efficiencies, and set a scalable template for system-wide adoption.

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Closing

CMS stands on the threshold of a new era, empowered to set the pace for digital transformation in healthcare. By embracing event-based data streaming architectures, open integration standards, and robust data governance, CMS can offset and eventually replace the patchwork of legacy systems with a living, resilient, and extensible platform. Confluent is poised to be a partner on this journey, offering a bridge from vision to reality—a future in which secure, real-time information powers better experiences and outcomes for every patient, clinician, and administrator in the healthcare ecosystem.

Further information on Confluent's Data Streaming Platform and the solutions we support can be found below:

https://www.confluent.io/data-streaming/
https://www.confluent.io/solutions/

Sincerely,
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